**25901279** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

## Fourth SEMESTER

## Discipline Specific Core Course (DSC) - MG4DSCFSC201 - FORENSIC PHYSICS

(2024 ADMISSION ONWARDS)

**Duration: 2 hours** **Maximum Marks: 70**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

One word answer Questions

Answer all questions

Each question carries **1** marks

1. Which compound contributes most to early strength in cement? [U] / [CO3]
2. Which method separates soil components by density? [U] / [CO1]
3. Poor binding in road layers indicates failure of which material? [U] / [CO3]
4. Name the process of reducing noise in a speech signal . [K] / [CO4]
5. Differentiate speaker variability from disguise. [U] / [CO5]
6. The factor affected most by background noise during analysis is. [An] / [CO4]
7. Describe the property that changes with different speakers. [U] / [CO5]
8. What is the first point of collision at a road accident scene called? [U] / [CO3]

9. The comparison of unknown and known speech samples is termed. [U] / [CO4]

10. What is the main constituent of soda-lime glass? [K] / [CO1]

**[1x10 = 10]**

## Part B

Short Answer Questions

Answer any **5** questions

Each question carries **3** marks

11. Describe the procedure for sealing and labeling physical evidence. [U] / [CO2]

12. Explain why glass fragments travel opposite to the direction of force. [K] / [CO1]

13. Explain the role of weathering in soil formation. [U] / [CO1]

14. Explain forensic examination of electrical installations [U] / [CO3]

15. Describe the process of speech production. [U] / [CO5]

16. Describe the procedure for transcription of disputed utterances. [S] / [CO4]

17. Write admissibility of voice evidence to court of law [A] / [CO5]

**[3x5 = 15]**

## Part C

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

18. Explain interpretation of soil evidence with examples. [U] / [CO1]

19. Explain the evidentiary value of glass fragments with suitable examples. [K] / [CO1]

20. Explain the mechanism of glass fracture and its forensic importance. [K] / [CO1]

21. Analyse the role of feature extraction in forensic speaker comparison. [An] / [CO5]

22. Describe the principles of pattern recognition and comparison in speech analysis. [U] / [CO5]

23. Discuss the legal aspects related to forensic speaker recognition and audio evidence. [U] / [CO4]

24. Analyze cement concrete failure in structural collapse. [An] / [CO3]

**[6x5 = 30]**

## Part D

Essay questions

Answer any **1** questions

Each question carries **15** marks

25. Apply particle size distribution, turbidity and microscopic examination in soil comparison. [A] / [CO1]

26. Classify the basic factors of sound in speech acoustics and examine their significance in speech analysis [A] / [CO5]

**[15x1 = 15]**